



Release of microplastics from polymeric ultrafiltration membrane system for drinking water treatment under different operating conditions

Thitiwut Maliwan , Jiangyong Hu ^{*}

Department of Civil & Environmental Engineering, National University of Singapore, 1 Engineering Drive 2, 117576, Singapore

ARTICLE INFO

Keywords:

Microplastics
Ultrafiltration
Polymeric membrane
Membrane cleaning
Drinking water

ABSTRACT

Drinking water has emerged as an important route for microplastics (MPs) to enter the human body, prompting concerns about their adverse health impacts. Membrane filtration technology is widely recognized as an effective treatment solution for combating MP pollution in water. However, recent research disputes that polymeric membrane systems may serve as additional sources of MPs in drinking water. The aim of this research is to investigate MP release from ultrafiltration membrane systems under different operating conditions by providing concrete evidence, identifying the operational factors contributing to the release, and elucidating the underlying possible mechanisms. Two key pieces of evidence were found to support the assertion that MPs were released from membrane systems, i.e., negative removal efficiency and an alteration in MP compositions observed between feed and permeate samples. Surprisingly, the MPs released from the membrane system originated not only from the membrane material and its additives but also from plastic-made equipment and even the other polymers used in the system. Overall results reveal that destructive activities such as shear stress, mechanical abrasion, and chemical oxidation processes, along with the carrying of MPs from external sources, are identified as potential mechanisms driving the concentration increase and polymer composition shift of MPs in permeate water. This study enhances an understanding of MP pollution in drinking water caused by membrane technology, potentially spurring the development of mitigation strategies for this issue.

1. Introduction

The current surge in global concern over emerging environmental contaminants, particularly microplastic (MP) pollution, has propelled it into the forefront of scientific research and public awareness (Qin et al., 2020). MPs, plastic particles with a size smaller than 5 mm, are pervasive and pose threats across various environmental media, including freshwater, seawater, air, and even subsurface layers like soil and groundwater (Boyle and Örmeci, 2020; Duis and Coors, 2016; Wu et al., 2020; Zhao et al., 2023). These plastic particles exhibit distinct features, such as being lightweight, durable, and resistant to degradation (Hu and Hu, 2024; Tian et al., 2023). Uncertain toxicity, their role as carriers of other toxic substances, and acting as hosts for pathogens and antibiotic resistant bacteria further emphasize their adverse effects (Bostan et al., 2023; Li et al., 2023; Pham et al., 2021; Rafa et al., 2024). Furthermore, numerous studies have linked the massive production of plastics over decades with waste mismanagement and the deterioration of these plastics through environmental processes, i.e., chemical oxidation, photo-oxidation, mechanical abrasion, and biodegradation, highlighting

the complicated fate and transport of MPs in the global environmental cycle (Browne et al., 2011; Hale et al., 2020; Hu et al., 2023).

The ubiquity of MPs in diverse environmental media exposes humans and various living organisms to these particles through several routes. Researchers have extensively studied the ecotoxicity and potential risks associated with MPs in some species of fish, crustaceans, mussels, algae, and even in rotifers (Ladeia Ramos et al., 2024; Nugnes et al., 2022). The daily ingestion or diffusion of MPs by these aquatic organisms sets off cascading impacts through biomagnification and bioaccumulation, affecting the entire food chains of the ecosystem. Beyond aquatic life, these minuscule plastic particles can enter the human body through the consumption of food and water, direct inhalation, and dermal penetration (Sun and Wang, 2023). Prior studies found MP particles, measuring up to 469 µm, in samples taken from the human heart and its surrounding tissue (Yang et al., 2023b). Furthermore, researchers have detected MPs in samples of human blood and lung tissue (Ladeia Ramos et al., 2024). Exacerbating these concerns is the incomplete understanding of the full extent of MP implications on human health, attributed to their uncertain toxicity and their ability as carriers of toxic

^{*} Corresponding author.

E-mail address: ceehujy@nus.edu.sg (J. Hu).

<https://doi.org/10.1016/j.watres.2024.123047>

Received 26 October 2024; Received in revised form 4 December 2024; Accepted 24 December 2024

Available online 25 December 2024

0043-1354/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.